Mill-duty brake - design brief

AI BRAKING / AB-AIST / C0 / 2026-09-07

A 12-inch drum layout explores a mill-duty envelope using the same modelling primitives. No AISE/AIST interchangeability, duty classification or compliance is asserted.

TARGETS (not validated)

Drum diameter: 304.8 mm / 12 in. Study geometry

Shoe width: 100 mm. Not a standard interface

Study holding torque: 500 N·m. Hypothesis; duty not received

Mounting pattern: Unreleased. Customer interface drawing required

DESIGN DECISIONS

Keep a separate interface family instead of relabelling the DIN-oriented concept.

Buy actuation until the severe-duty force/time envelope is known.

Prioritize access for inspection and replacement in dirty environments.

CANDIDATE MATERIALS

S355J2+N candidate structure with environment-specific coating.

Rotor and lining pair require mill-duty temperature and contamination assessment.

Assembly and engineering gates

ASSEMBLY INTENT

1. Obtain machine interface dimensions and applicable standard revision.
2. Review base, pivot and thruster positions against that interface.
3. Proceed with the drum-platform assembly sequence only after separate design approval.

OPEN DESIGN RISKS

1. All mounting dimensions are original placeholders, not an AIST drawing.
2. High cycle duty, hot surroundings and corrosion require a separate load spectrum.

VALIDATION REQUIRED

1. Interface metrology and full duty-cycle definition.
2. Contaminated/environmental endurance and thermal testing.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

1. Advisory position / wear sensing with replaceable brackets; no AI-controlled brake release.
2. Record supplier sensor range, supply, connector, EMC, ingress and temperature limits before selection.
3. Define independent machine safety interface and diagnostic fault response with the responsible controls engineer.

Assembly reference / numbered components

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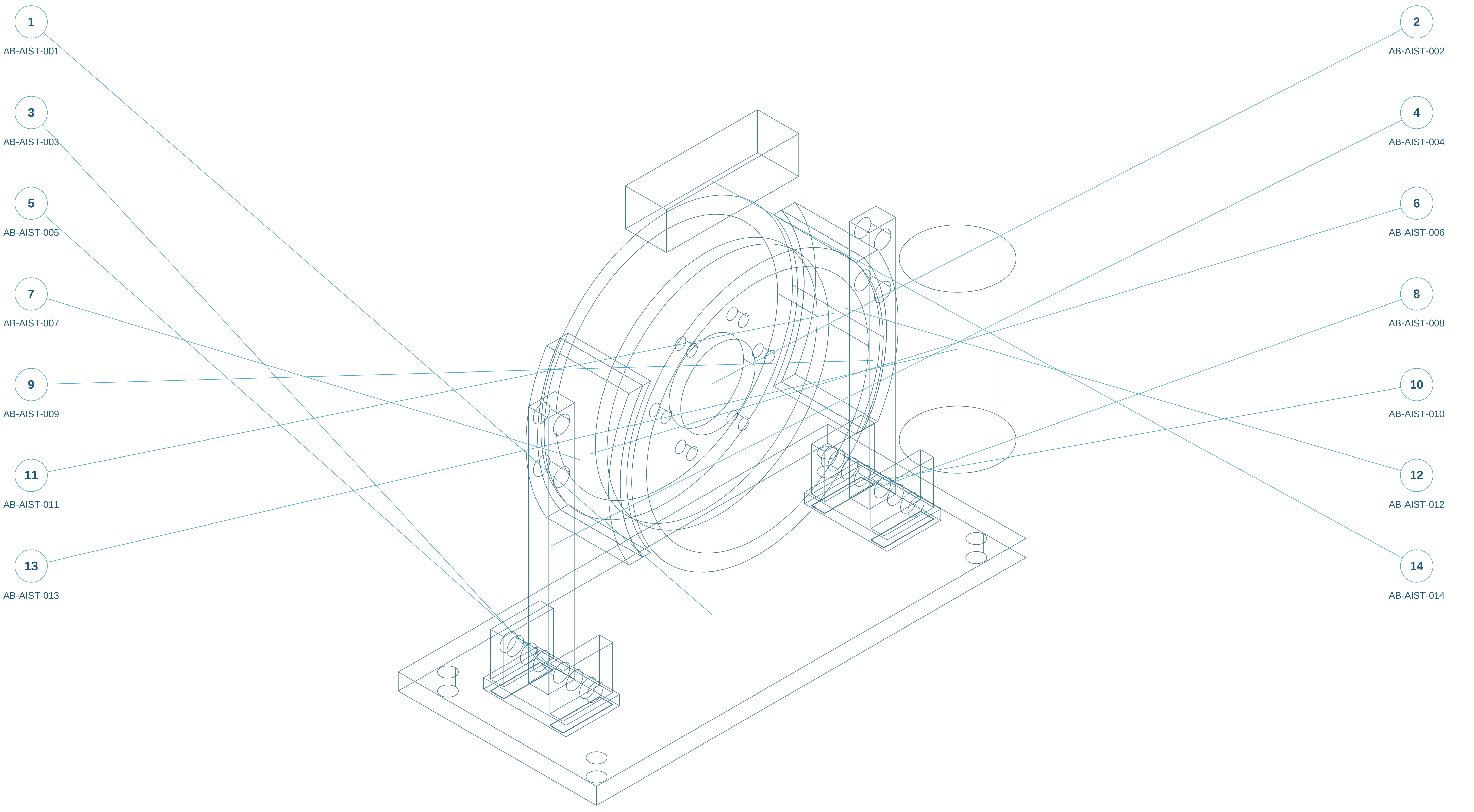

Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

AB-AIST-001 | Four-hole base | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-AIST-002 | Study brake drum | make-concept

C45 prototype candidate; friction/thermal qualification required; Saw/laser blank; CNC critical faces and bores

AB-AIST-003 | Pivot clevis -1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-AIST-004 | Lever -1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-AIST-005 | Lower pivot pin -1 | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Turn/grind; specify retention after review

AB-AIST-006 | Lining -1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy friction lining / machine carrier to approved retention drawing

AB-AIST-007 | Shoe carrier -1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Buy friction lining / machine carrier to approved retention drawing

AB-AIST-008 | Pivot clevis 1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-AIST-009 | Lever 1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Saw/laser blank; CNC critical faces and bores

AB-AIST-010 | Lower pivot pin 1 | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Turn/grind; specify retention after review

AB-AIST-011 | Lining 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy friction lining / machine carrier to approved retention drawing

AB-AIST-012 | Shoe carrier 1 | make-concept

S355J2+N candidate; condition and design properties unapproved; Buy friction lining / machine carrier to approved retention drawing

AB-AIST-013 | Thruster reserved space | bought-envelope

Supplier-controlled; illustrative reserved envelope; Buy selected thruster; replace with vendor STEP

AB-AIST-014 | Spring/equalizer reserved space | interface-envelope

Supplier-controlled; illustrative reserved envelope; Engineer spring/linkage after kinematics

AB-AIST-001 / Four-hole base

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S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 520.00 / 240.00 / 20.00 mm

520 x 240 x 20; 4 x diameter 18 THRU.

Hole centres X +/-230, Y +/-90 from part origin.

Machine base datum; mounting preload unassigned.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

FRONT X-Z

SIDE Y-Z

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-AIST-002 / Study brake drum

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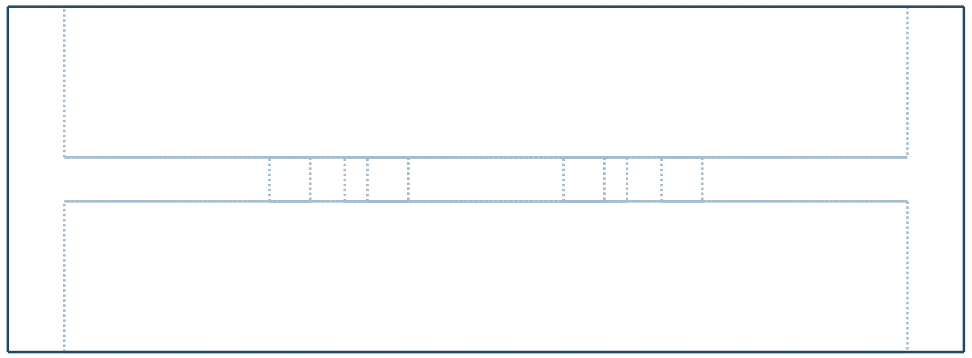
TOP X-Y



C45 prototype candidate; friction/thermal qualification required

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 304.80 / 110.00 / 304.80 mm

OD 304.8; friction width 100 within 110 mm axial envelope.

Web bore 90; 6 x diameter 13 on PCD 125.

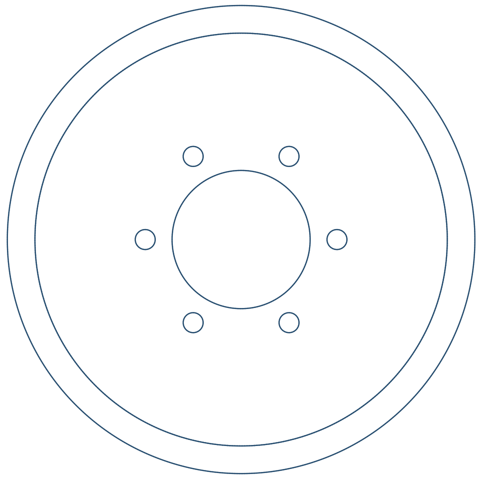
Shaft hub and retention not detailed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

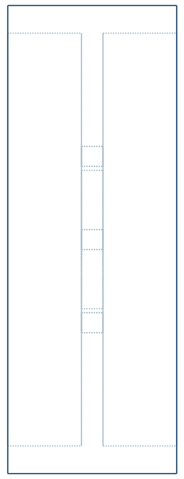
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-AIST-003 / Pivot clevis -1

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S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 65.00 / 100.00 / 65.00 mm

Clevis pin bore 20; nominal ear gap 56 mm.

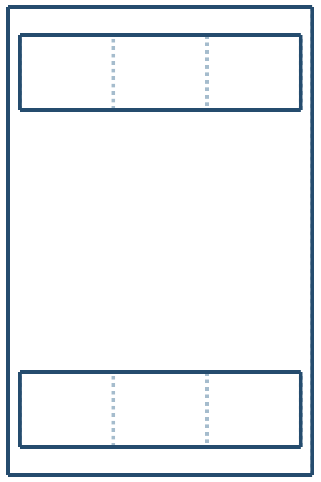
Foot 4 x diameter 9 at (+/-20,+/-35); base mating holes pending.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

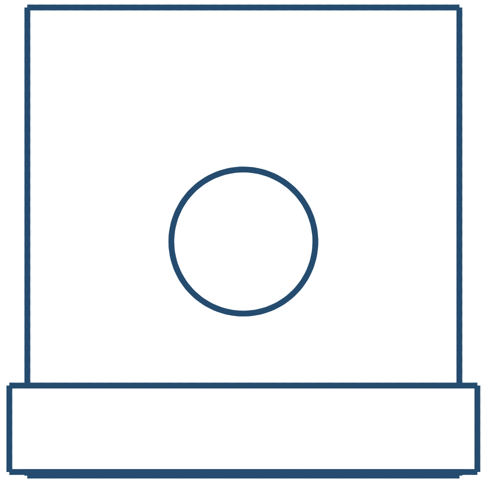
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

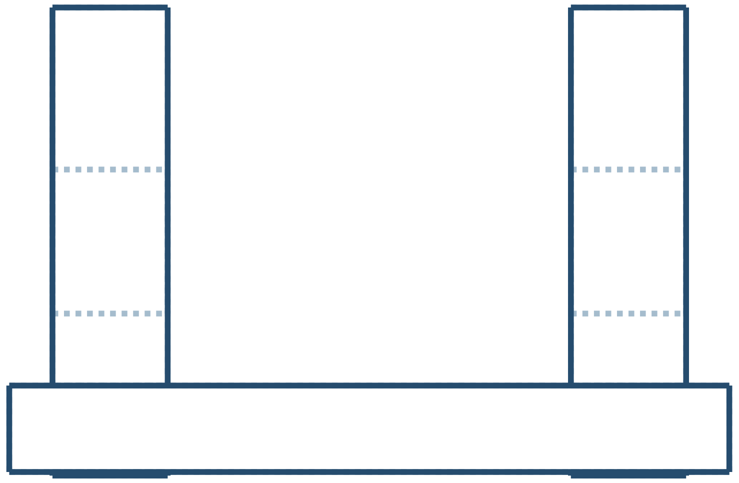
TOP X-Y



FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-AIST-004 / Lever -1

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S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 32.00 / 24.00 / 290.00 mm

32 x 24 section; pin centres Z 0, 205, 260 in local coordinates.

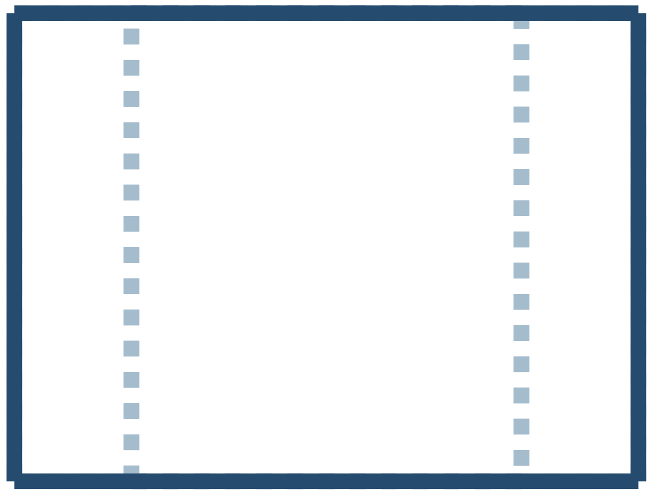
Nominal pin bore 20; bush fit not detailed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y



FRONT X-Z

SIDE Y-Z

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-AIST-005 / Lower pivot pin -1

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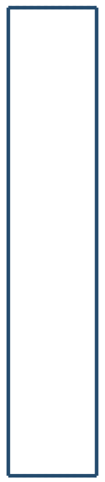
TOP X-Y



42CrMo4 candidate; heat treatment / fit unapproved

Turn/grind; specify retention after review

Overall X/Y/Z bounds: 19.80 / 105.00 / 19.80 mm

Diameter 19.8 study clearance in nominal 20 bore.

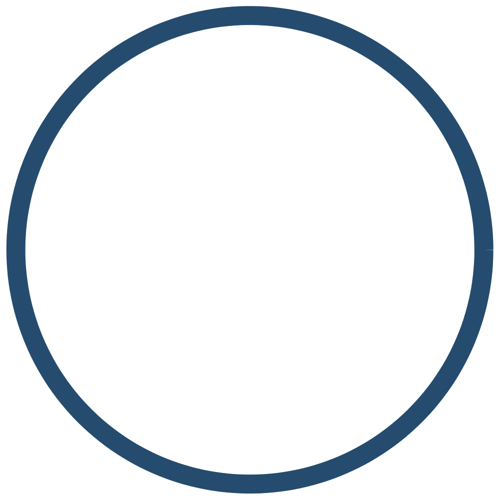
No final fit, hardness or axial retainer design.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z

AB-AIST-006 / Lining -1

AI BRAKING / AB-AIST / C0 / 2026-09-07

Supplier-controlled; illustrative reserved envelope

Buy friction lining / machine carrier to approved retention drawing

Overall X/Y/Z bounds: 37.18 / 100.00 / 180.00 mm

1 mm radial study gap at drum; not a release-clearance specification.

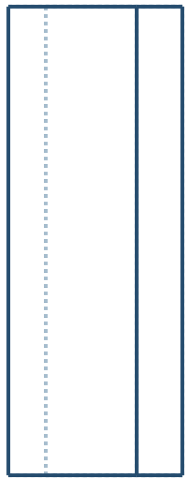
Carrier pivots and lining retention remain to be designed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

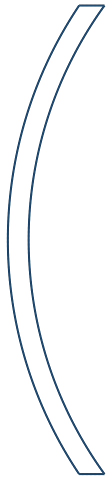
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

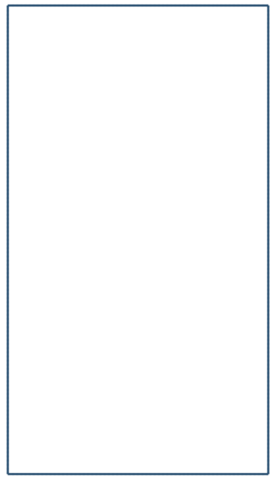
TOP X-Y



FRONT X-Z



SIDE Y-Z



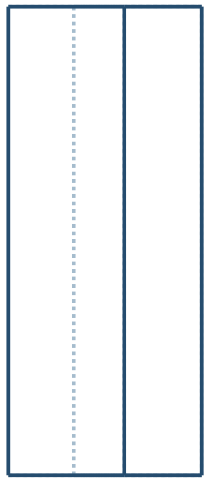
ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-AIST-007 / Shoe carrier -1

AI BRAKING / AB-AIST / C0 / 2026-09-07

TOP X-Y



S355J2+N candidate; condition and design properties unapproved

Buy friction lining / machine carrier to approved retention drawing

Overall X/Y/Z bounds: 41.42 / 100.00 / 180.00 mm

1 mm radial study gap at drum; not a release-clearance specification.

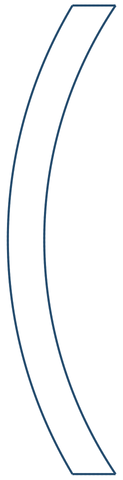
Carrier pivots and lining retention remain to be designed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

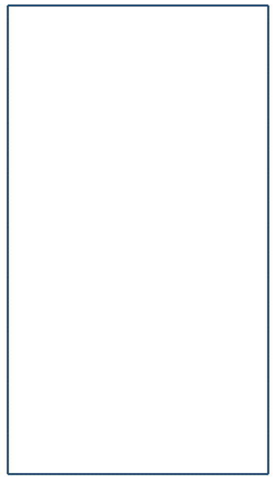
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 65.00 / 100.00 / 65.00 mm

Clevis pin bore 20; nominal ear gap 56 mm.

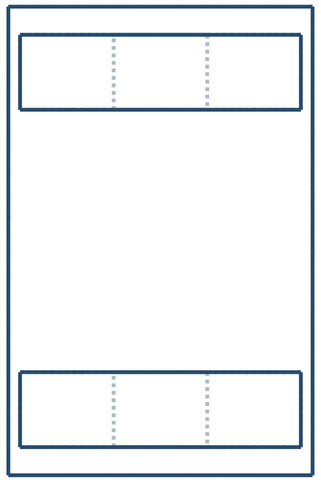
Foot 4 x diameter 9 at (+/-20,+/-35); base mating holes pending.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

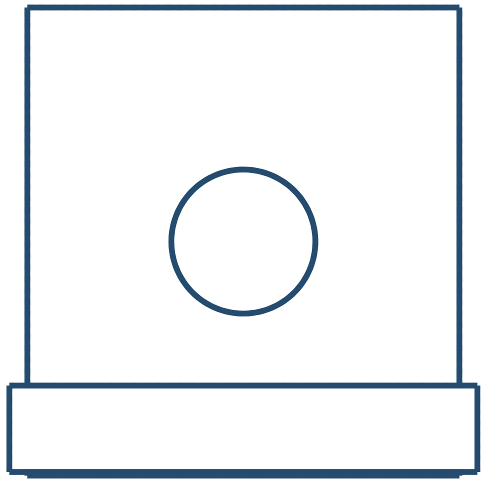
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

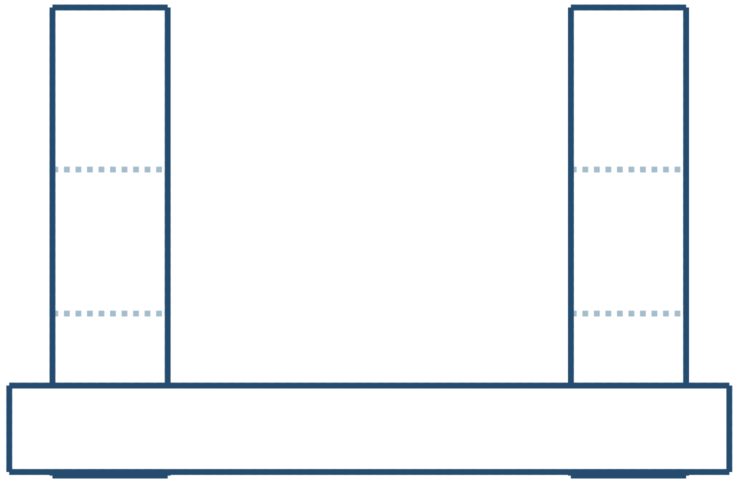
TOP X-Y



FRONT X-Z



SIDE Y-Z



AB-AIST-009 / Lever 1

AI BRAKING / AB-AIST / C0 / 2026-09-07

S355J2+N candidate; condition and design properties unapproved

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 32.00 / 24.00 / 290.00 mm

32 x 24 section; pin centres Z 0, 205, 260 in local coordinates.

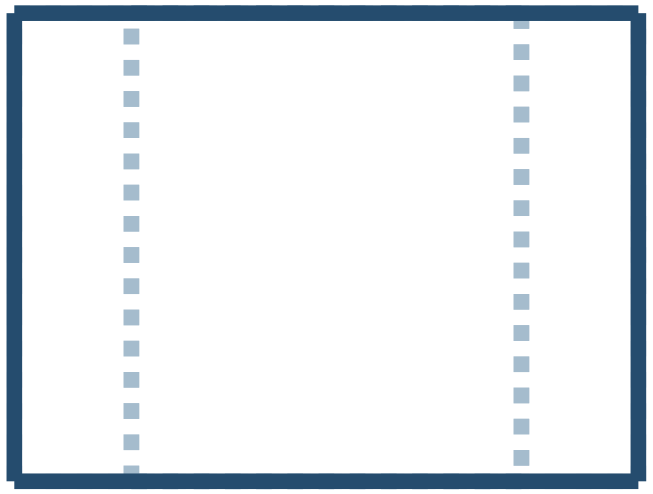
Nominal pin bore 20; bush fit not detailed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y



FRONT X-Z

SIDE Y-Z

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-AIST-010 / Lower pivot pin 1

AI BRAKING / AB-AIST / C0 / 2026-09-07

42CrMo4 candidate; heat treatment / fit unapproved

Turn/grind; specify retention after review

Overall X/Y/Z bounds: 19.80 / 105.00 / 19.80 mm

Diameter 19.8 study clearance in nominal 20 bore.

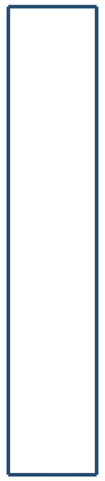
No final fit, hardness or axial retainer design.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

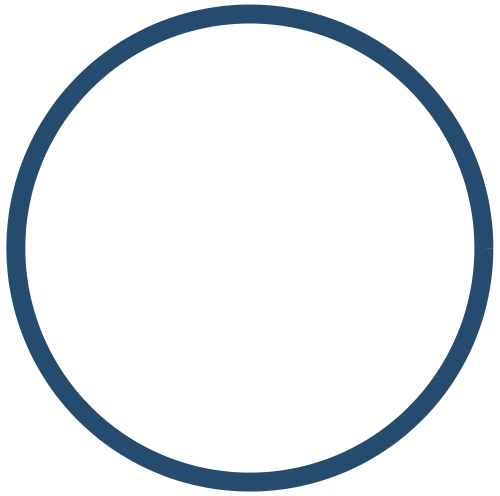
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y



FRONT X-Z



SIDE Y-Z

AB-AIST-011 / Lining 1

AI BRAKING / AB-AIST / C0 / 2026-09-07

Supplier-controlled; illustrative reserved envelope

Buy friction lining / machine carrier to approved retention drawing

Overall X/Y/Z bounds: 37.18 / 100.00 / 180.00 mm

1 mm radial study gap at drum; not a release-clearance specification.

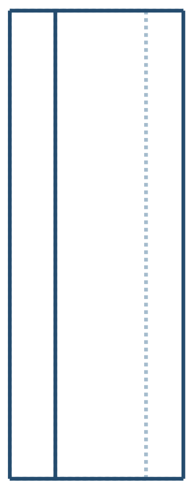
Carrier pivots and lining retention remain to be designed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

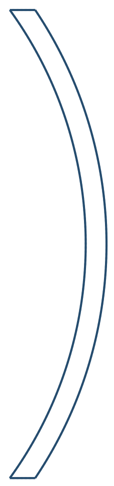
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

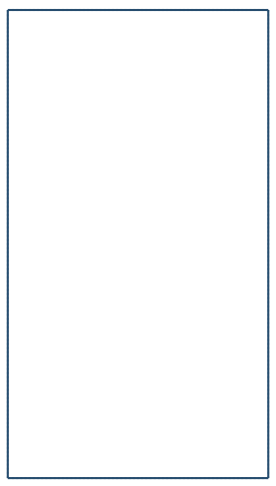
TOP X-Y



FRONT X-Z



SIDE Y-Z



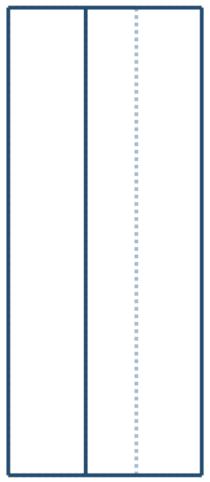
ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-AIST-012 / Shoe carrier 1

AI BRAKING / AB-AIST / C0 / 2026-09-07

TOP X-Y



S355J2+N candidate; condition and design properties unapproved

Buy friction lining / machine carrier to approved retention drawing

Overall X/Y/Z bounds: 41.42 / 100.00 / 180.00 mm

1 mm radial study gap at drum; not a release-clearance specification.

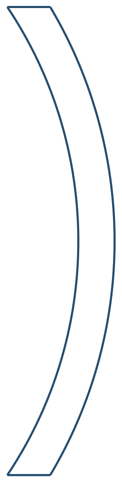
Carrier pivots and lining retention remain to be designed.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

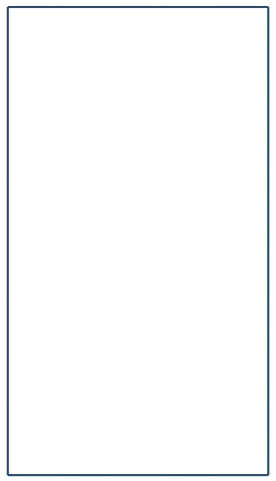
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

AB-AIST-013 / Thruster reserved space

AI BRAKING / AB-AIST / C0 / 2026-09-07

Supplier-controlled; illustrative reserved envelope

Buy selected thruster; replace with vendor STEP

Overall X/Y/Z bounds: 100.00 / 100.00 / 190.00 mm

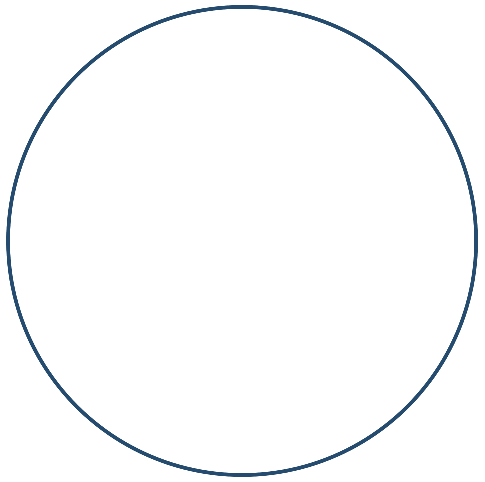
Nonfunctional body envelope; linkage not included.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

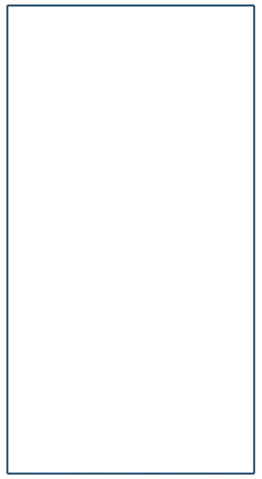
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

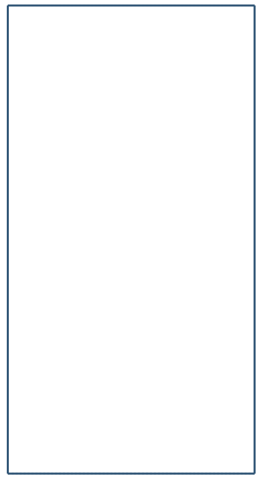
TOP X-Y



FRONT X-Z



SIDE Y-Z



AB-AIST-014 / Spring/equalizer reserved space

AI BRAKING / AB-AIST / C0 / 2026-09-07

Supplier-controlled; illustrative reserved envelope

Engineer spring/linkage after kinematics

Overall X/Y/Z bounds: 160.00 / 50.00 / 45.00 mm

Reserved volume only; no spring, force path or release motion solved.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

FRONT X-Z

SIDE Y-Z